

A Novel Partial Key Exposure Attack on Common Prime RSA

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Abstract. We propose a novel partial key exposure attack on common prime RSA by leveraging lattice-based techniques. In common prime RSA, the primes p and q are defined as $p = 2ga + 1$ and $q = 2gb + 1$ for a common prime g . Recently, Zheng introduced the first partial key exposure attack on this scheme; however, it is limited to instances where $g > N^{1/4}$. In contrast, our work investigates deeper into partial key exposure attacks by presenting a unified generic case that targets one consecutive unknown block of the private key. By employing a lattice-based solving strategy for trivariate integer polynomials, we can effectively identify additional weak private keys that are vulnerable to partial exposure. Extensive numerical experiments validate the correctness and practicality of our proposed attack on common prime RSA.

Keywords: Common Prime RSA · Cryptanalysis · Lattice · Partial Key Exposure Attack · Trivariate Integer Polynomial

1 Introduction

The RSA cryptosystem, originally introduced by Rivest, Shamir, and Adleman [24], remains one of the most prominent public key encryption schemes. To reduce the high computational overhead associated with the original RSA, several variants have been developed, including CRT-RSA [22], multi-prime RSA [6], prime-power RSA [27], and common prime RSA [11]. In this paper, we focus on partial key exposure attacks against the common prime RSA variant by employing a lattice-based solving strategy, which serves as an effective tool to assess cryptanalytic vulnerabilities in RSA and its variants.

Consider an RSA instance with a public key (N, e) and a private key (p, q, d) , where the modulus is defined as $N = pq$ with p and q being balanced primes. In the original RSA scheme [24], the exponents e and d are selected as mutual inverses modulo Euler's totient function $\varphi(N) = (p - 1)(q - 1)$. However, it has become customary to define these exponents modulo Carmichael's lambda function [8], i.e.,

$$\lambda(N) = \text{lcm}(p - 1, q - 1),$$

which represents the least common multiple of $p - 1$ and $q - 1$. In the case of common prime RSA, a variant first suggested by Wiener [31] and later refined by Hinek [11], the balanced primes p and q possess a special structure that enhances resistance against small private key attacks.

We now delve into the mathematical underpinnings of common prime RSA as described in [11]. In Hinek's design, the balanced primes are defined as

$$p = 2ga + 1, \quad q = 2gb + 1, \quad (1)$$

where g is a common prime and a, b are positive integers. These primes p, q are subject to two constraints: $\gcd(a, b) = 1$, which ensures that $\gcd(p-1, q-1) = 2g$, and the requirement that $h = 2gab + a + b$ is prime, so that $(pq - 1)/2 = gh$ is a semiprime with a bit-length comparable to that of the RSA modulus N . A modified version of the common prime generation algorithm from [11] is provided in Algorithm 1 in Appendix A.

Within the common prime RSA framework, the public and private exponents e and d are defined modulo

$$\lambda(N) = \lambda(pq) = \text{lcm}(p-1, q-1) = \text{lcm}(2ga, 2gb) = 2gab.$$

According to Hinek's construction, the modulus can be expressed as

$$N = pq = (2ga + 1)(2gb + 1) = 2g(2gab + a + b) + 1 = 2gh + 1. \quad (2)$$

Moreover, the key congruence

$$ed \equiv 1 \pmod{2gab}, \quad (3)$$

can be equivalently written as

$$ed = 2gabk + 1, \quad (4)$$

where k is an unknown positive integer relatively prime to $2g$.

Throughout this paper, we use the symbol $\simeq$ to indicate asymptotic equivalence with respect to a certain power of the RSA modulus N . In particular, we assume that the large prime satisfies $g \simeq N^\gamma$ and that the private exponent obeys $d \simeq N^\delta$. Since balanced primes yield $\gcd(p-1, q-1) = 2g$, it follows that $0 < \gamma < 1/2$. Furthermore, by assuming that the bit-length of e is approximately that of N/g , we have $e \simeq N^{1-\gamma}$.

The security of common prime RSA has been extensively analyzed in the literature [11, 14, 17, 21, 26, 35]. Notably, partial key exposure attacks, where an adversary obtains a fraction of the private key bits, are particularly pertinent in resource-constrained environments such as the Internet of Things [21]. These attacks exploit side-channel information acquired through methods including cold boot attacks [10] and various other side-channel analysis techniques [15, 23, 25]. Therefore, a rigorous investigation of partial key exposure attacks on common prime RSA is of considerable importance.

Partial key exposure attacks on RSA, which rely on the leakage of partial information about the private key, were initially introduced by Boneh et al. [5]. Such attacks typically exploit the knowledge of either the most significant bits (MSBs) or the least significant bits (LSBs) of the private exponent d , information that may be obtained via side-channel attacks. Subsequently, Blömer and May [2] refined these techniques using Coppersmith’s lattice-based techniques [7], demonstrating that RSA becomes vulnerable even for larger public exponents when some private key bits are exposed. Ernst et al. [9] further proposed attacks applicable to full-size exponents (i.e., $e \simeq N$ or $d \simeq N$) based on three theorems under a common heuristic assumption. The most effective partial key exposure attack on standard RSA to date was proposed by Takayasu and Kunihiro [29], which achieves Boneh-Durfee’s bound [4] for small private key attacks. In addition, similar attack techniques have been extended to other RSA variants [13, 20, 32]. However, research focusing on partial key exposure attacks for the common prime RSA scheme is rare, with Zheng’s work [33, 34] being an exception.

1.1 Related Work

Recently, Zheng [33, 34] presented the first dedicated investigation of partial key exposure attacks on common prime RSA. His study begins by the congruence $ed \equiv 1 \pmod{2gab}$. To address the small private key attack scenario on common prime RSA, his method focuses on solving the congruence $ex - 1 \equiv 0 \pmod{g}$, where $x = d$ is treated as the small root.

To extend the analysis to partial key exposure attacks, Zheng formalized several attack scenarios. These include cases where the MSBs of the private key, the LSBs, or both are leaked. Accordingly, the attack scenarios are classified into three cases: the MSB case, the LSB case, and the MSB-LSB case. In each scenario, the private key d assumes a different form based on the leaked information. By substituting the corresponding form of d into the congruence $ex - 1 \equiv 0 \pmod{g}$ and applying a modular inverse, a new modular univariate linear equation is derived for each scenario. To solve these equations, Zheng employs the lattice-based method introduced by Coppersmith [7].

Furthermore, an auxiliary modular univariate linear equation can be derived from the identity $(p-1)(q-1) = 2ga \cdot 2gb = 4g^2ab$, which implies $pq - p - q + 1 \equiv 0 \pmod{g^2}$. Thus, one seeks to solve $x + N + 1 \equiv 0 \pmod{g^2}$, with $x = -(p + q)$ as the small root. In this way, the problem reduces to solving two simultaneous modular univariate linear equations. The following proposition summarizes the first demonstration of the MSB-LSB case partial key exposure attack on common prime RSA.

Proposition 1 ([34]). *Let $N = pq$ be an ℓ -bit common prime RSA modulus with balanced primes p and q of equal bit-length. Let $e \simeq N^{1-\gamma}$ and $d \simeq N^\delta$ be the public and private exponents satisfying*

$$ed \equiv 1 \pmod{\text{lcm}(p-1, q-1)},$$

with $\gcd(p-1, q-1) = 2g$ for a common prime $g \simeq N^\gamma$. Suppose an approximation of d is available, where the most significant bits d_M form a $(\delta_M \ell)$ -bit block and the least significant bits d_L form a $(\delta_L \ell)$ -bit block, such that

$$d = d_M M + \bar{d} L + d_L,$$

with $M = 2^{(\delta - \delta_M)\ell}$, $L = 2^{\delta_L \ell}$, and the unknown component $\bar{d}$ satisfying $|\bar{d}| \leq N^{\delta - \delta_M - \delta_L}$. Then, N can be efficiently factored in time polynomial in ℓ if

$$\delta_M + \delta_L < \delta < 4\gamma^3 + \delta_M + \delta_L, \quad \frac{1}{4} \leq \gamma < \frac{1}{2}.$$

The aforementioned work explores a novel domain in which partial leakage of private key bits is assumed. Zheng [34] proposed three distinct attacks corresponding to the MSB, LSB, and MSB-LSB cases. His results demonstrate new advancements in assessing the security of common prime RSA relative to previous small private key attacks. Moreover, Zheng noted that several enhancements, such as the adoption of more efficient lattice-based strategies, could further improve the primary results. This observation leaves ample room for future work to generalize partial key exposure attacks on common prime RSA.

1.2 Our Contribution

Although Zheng’s work represents a seminal contribution to partial key exposure attacks on common prime RSA, it is subject to the limitation that $1/4 \leq \gamma < 1/2$. Hence, there is a need to develop attacks for scenarios where the prime g is smaller (i.e., $\gamma < 1/4$). Inspired by the small private key attack on common prime RSA [14], we propose a novel partial key exposure attack based on solving trivariate integer polynomial equations, where the desired root is related to the secret parameters inherent in common prime RSA. Moreover, in contrast to the three separate scenarios presented in [34], our approach unifies them into a generic case that considers only one consecutive unknown block in the private key.

Our main contributions are summarized as follows:

- We extend the seminal partial-key-leakage attack on common-prime RSA [34] from the regime $1/4 \leq \gamma < 1/2$ down to *any* $\gamma < 1/2$, in particular covering $\gamma < 1/4$.
- We introduce a generic trivariate-polynomial based formulation that simultaneously handles MSB, LSB, and MSB-LSB exposures of the private key.
- We derive attack bounds on the maximum exploitable exponent size δ as a function of $(\gamma, \delta_M, \delta_L)$, improving previous limits.
- We validate the proposed attack in **SageMath** with 1024-bit and 2048-bit modulus experiments and publish our detailed implementation under an open-source license on **GitHub**[‡].

[‡] https://github.com/MengceZheng/CPRSA_PKEA

1.3 Organization

The remainder of this paper is organized as follows. In Section 2 we introduce key mathematical lemmas and fundamental facts pertinent to the lattice-based solving strategy. Section 3 details our new partial key exposure attack on common prime RSA. In Section 4 we present numerical experimental results to substantiate the correctness and effectiveness of the proposed attack. Finally, Section 5 concludes the paper.

2 Preliminaries

This section reviews the fundamental concepts and techniques employed in our cryptanalysis, including lattice reduction, mostly the LLL algorithm [16], and Coppersmith's lattice-based method [7], which was later refined via Howgrave-Graham's lemma [12]. For a more comprehensive understanding, readers are referred to [18, 19, 36]. We begin by defining a lattice as a discrete additive subgroup of $\mathbb{R}^n$.

Definition 1. A lattice Λ is the set of all integer linear combinations of ω linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_\omega \in \mathbb{R}^n$, i.e.,

$$\Lambda = \left\{ \sum_{i=1}^{\omega} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}.$$

The *determinant* of a lattice, denoted $\det(\Lambda)$, is defined as $\sqrt{\det(BB^\top)}$, where B is the basis matrix whose rows are the basis vectors $\mathbf{b}_1, \dots, \mathbf{b}_\omega$ and $B^\top$ denotes its transpose. In the case of a full-rank lattice (i.e., $\omega = n$), the determinant simplifies to $\det(\Lambda) = |\det(B)|$. The integers n and ω are referred to as the *dimension* and the *rank* of the lattice, respectively.

The LLL algorithm [16] is a widely used tool for efficiently obtaining an approximately short basis for a lattice. As shown in [18], the LLL algorithm produces a reduced basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\omega)$ satisfying the following bound.

Lemma 1 ([16, 18]). Let Λ be an ω -dimensional lattice. The LLL algorithm returns a reduced basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\omega)$ such that, for each $1 \leq i \leq \omega$,

$$\|\mathbf{v}_1\|, \dots, \|\mathbf{v}_i\| \leq 2^{\frac{\omega(\omega-1)}{4(\omega+1-i)}} \det(\Lambda)^{\frac{1}{\omega+1-i}}.$$

The running time of the LLL algorithm is polynomial in ω and in the logarithm of the maximal input vector component.

Howgrave-Graham's lemma [12] provides a criterion to decide when a root of a modular polynomial equation is also a root over the integers. Consider an integer polynomial

$$g(x_1, \dots, x_n) = \sum a_{i_1, \dots, i_n} x_1^{i_1} \cdots x_n^{i_n}$$

with the norm defined by

$$\|g(x_1, \dots, x_n)\| := \sqrt{\sum |a_{i_1, \dots, i_n}|^2}.$$

The following lemma is instrumental in our analysis.

Lemma 2 ([12]). *Let $g(x_1, \dots, x_n) \in \mathbb{Z}[x_1, \dots, x_n]$ be an integer polynomial comprising at most ω monomials. Let $R, X_1, \dots, X_n$ be given positive integers. If there exists a tuple $(x_1^*, \dots, x_n^*)$ with $|x_i^*| \leq X_i$ for $1 \leq i \leq n$ such that*

$$g(x_1^*, \dots, x_n^*) \equiv 0 \pmod{R},$$

and if

$$\|g(x_1 X_1, \dots, x_n X_n)\| < \frac{R}{\sqrt{\omega}},$$

then $g(x_1^*, \dots, x_n^*) = 0$ over the integers.

By combining the output of the LLL algorithm with Howgrave-Graham's lemma, one can effectively solve modular or integer polynomial equations. In practice, if one computes the first ℓ reduced vectors of the lattice, successful application of the lattice-based solving strategy is achieved provided that

$$2^{\frac{\omega(\omega-1)}{4(\omega+1-\ell)}} \det(\Lambda)^{\frac{1}{\omega+1-\ell}} < \frac{R}{\sqrt{\omega}}.$$

This condition can be rearranged as

$$\det(\Lambda) < R^{\omega+1-\ell} 2^{-\frac{\omega(\omega-1)}{4}} \omega^{-\frac{\omega+1-\ell}{2}}.$$

Since $\ell < \omega \ll R$, this further implies, up to a small error term, that $\det(\Lambda) < R^{\omega-\epsilon}$, which asymptotically leads to the condition

$$\det(\Lambda) < R^{\omega}. \quad (5)$$

This inequality forms the basis for effectively solving the targeted modular or integer polynomial equations.

A critical aspect of the lattice-based solving strategy is the construction of an appropriate lattice. Several works (e.g., [3, 14, 28]) have focused on designing lattices with optimized solving conditions. In our approach, we adopt the strategy proposed by Jochemsz-May [14], which constructs a lattice with a triangular basis matrix so that the determinant is easily computed as the product of its diagonal elements.

In our setting, we seek a small root of a trivariate integer polynomial of the form

$$f(x, y, z) = a_1 x^2 + a_2 xy + a_3 xz + a_4 yz + a_5 x + a_6 y + a_7 z + a_0.$$

We now describe in detail the lattice-based strategy to solve for the small root (x^*, y^*, z^*) of $f(x, y, z)$.

Polynomial Target: Suppose $f(x, y, z)$ is an irreducible trivariate polynomial with the property that $f(x^*, y^*, z^*) = 0$ for some unknown triple (x^*, y^*, z^*) satisfying the bounds $|x^*| \leq X$, $|y^*| \leq Y$, and $|z^*| \leq Z$. Define

$$U = \|f(xX, yY, zZ)\|_\infty, \quad (6)$$

which represents the maximum absolute value of the coefficients when the variables are scaled by their bounds. An additional integer parameter R is then introduced as a useful modulus. If the constant term $a_0 \neq 1$, one may rescale the polynomial via $f(x, y, z) \leftarrow a_0^{-1}f(x, y, z) \pmod{R}$ to ensure the constant term equal to 1.

Shift Construction: Using $f(x, y, z)$, one constructs a family of *shift polynomials* $g_{[i,j,k]}(x, y, z)$ such that each satisfies

$$g_{[i,j,k]}(x^*, y^*, z^*) \equiv 0 \pmod{R}.$$

A standard method is to define two monomial sets:

$$\begin{aligned} \mathcal{M}_\uparrow &= \bigcup_{0 \leq l \leq t} \{x^{i+l}y^jz^k : x^i y^j z^k \in \text{monomial}(f^{m-1})\}, \\ \mathcal{M}_\downarrow &= \bigcup_{0 \leq l \leq t} \{x^{i+l}y^jz^k : x^i y^j z^k \in \text{monomial}(f^m)\}, \end{aligned}$$

where m is a suitably chosen positive integer and $t \geq 0$ is an integer. These sets yield the corresponding index sets $\mathcal{I}_\uparrow$ and $\mathcal{I}_\downarrow$. Let μ_x , μ_y , and μ_z denote the maximum degrees in x , y , and z , respectively, appearing in $\mathcal{M}_\uparrow$. Then, set

$$R := U X^{\mu_x} Y^{\mu_y} Z^{\mu_z}.$$

The shift polynomials $g_{[i,j,k]}(x, y, z)$ are defined by

$$\begin{cases} x^i y^j z^k f(x, y, z) X^{\mu_x - i} Y^{\mu_y - j} Z^{\mu_z - k}, & \text{if } x^i y^j z^k \in \mathcal{M}_\uparrow, \\ x^i y^j z^k R, & \text{if } x^i y^j z^k \in \mathcal{M}_\downarrow \setminus \mathcal{M}_\uparrow. \end{cases}$$

By construction, each $g_{[i,j,k]}(x, y, z)$ vanishes modulo R at (x^*, y^*, z^*) .

Lattice Generation: Ordering the monomials and polynomials according to a chosen term order (e.g., ascending graded lexicographic order), one can form a lattice Λ by mapping the coefficient vectors of the scaled shift polynomials $g_{[i,j,k]}(xX, yY, zZ)$, for $[i, j, k] \in \mathcal{I}_\downarrow$, into row vectors $\mathbf{b}_1, \dots, \mathbf{b}_\omega$ of a basis matrix B , where $\omega = |\mathcal{M}_\downarrow|$. Under the chosen order, this matrix is full-rank and triangular, making the determinant easy to compute as the product of its diagonal entries.

Lattice Reduction: Applying the LLL algorithm to the basis matrix B , one obtains a sequence of reduced vectors $(\mathbf{v}_1, \dots, \mathbf{v}_\ell)$ (with $\ell \geq 2$). These vectors correspond to integer polynomials $h_i(x, y, z)$, each being a linear combination of the $g_{[i,j,k]}(x, y, z)$, and satisfying

$$h_i(x^*, y^*, z^*) \equiv 0 \pmod{R}, \quad 1 \leq i \leq \ell.$$

By Lemma 1 and Lemma 2, if the parameters involved satisfy the condition in (5), then the small root (x^*, y^*, z^*) is guaranteed to be a correct integer solution.

Root Extraction: Provided that at least two of the derived polynomials, say $h_1(x, y, z)$ and $h_2(x, y, z)$, are algebraically independent, the system

$$\mathcal{V} = \{f(x, y, z), h_1(x, y, z), h_2(x, y, z)\}$$

can be solved using techniques such as resultant or Gröbner basis computations [1] to recover the unique small root (x^*, y^*, z^*) . The overall running time is dominated by the lattice reduction and the subsequent root recovery, both of which can be performed in polynomial time relative to the input size.

It is important to note that the lattice-based solving strategy is heuristic in nature, as there is no absolute guarantee that the constructed integer polynomials will always be algebraically independent. In practice, however, it is commonly assumed in lattice-based attacks that the polynomials derived via the LLL algorithm exhibit algebraic independence.

Assumption 1. *The derived integer polynomials by the LLL reduction algorithm are algebraically independent, ensuring that their common root can be efficiently recovered.*

In most prior Coppersmith-style attacks on RSA variants (e.g. Boneh-Durfee [4], Jochemsz-May [14]) and in our own extensive experiments (see Section 4), the reduced vectors always yielded independent polynomials whose common small root was recovered with negligible failure rate.

3 Novel Partial Key Exposure Attack

We present a new partial key exposure attack on common prime RSA by applying the lattice-based solving strategy outlined in Section 2. Our main result is summarized in the following proposition.

Proposition 2. *Let $N = pq$ be an ℓ -bit common prime RSA modulus with balanced primes p and q of equal bit-length. Let $e \simeq N^{1-\gamma}$ and $d \simeq N^\delta$ be the public and private exponents satisfying*

$$ed \equiv 1 \pmod{\text{lcm}(p-1, q-1)},$$

with $\gcd(p-1, q-1) = 2g$ for a common prime $g \simeq N^\gamma$. Suppose an approximation of d is available, where the most significant bits d_M form a $(\delta_M \ell)$ -bit block and the least significant bits d_L form a $(\delta_L \ell)$ -bit block, such that

$$d = d_M M + \bar{d} L + d_L,$$

where $M = 2^{(\delta - \delta_M)\ell}$, $L = 2^{\delta_L\ell}$, and the unknown component $\bar{d}$ satisfies $|\bar{d}| \leq N^{\delta - \delta_M - \delta_L}$. Then, N can be efficiently factored in time polynomial in ℓ if

$$\begin{cases} \delta < \gamma + 1 - \frac{\sqrt{4(\gamma - \delta_M - \delta_L)^2 + 20(\gamma - \delta_M - \delta_L) + 13}}{4}, & \text{if } \gamma \leq \delta_M + \delta_L + \frac{3}{10}, \\ \delta < \frac{4\gamma + 7\delta_M + 7\delta_L + 1}{11}, & \text{if } \gamma > \delta_M + \delta_L + \frac{3}{10}. \end{cases}$$

Proof. Recall that common prime RSA satisfies the key equation $ed = 2gabbk + 1$, where k is an unknown positive integer. By substituting $2ga = p - 1$ and $2gb = q - 1$ into (4), we obtain

$$ed = (p - 1)bk + 1, \quad ed = (q - 1)ak + 1.$$

These equations imply

$$ed - 1 + bk = pbk, \quad ed - 1 + ak = qak.$$

Multiplying the two equations yields

$$(ed - 1 + bk)(ed - 1 + ak) = (pbk)(qak) = ak \cdot bk \cdot N.$$

Substitute the decomposition

$$d = d_M M + \bar{d} L + d_L$$

and treat $\bar{d}$, ak , and bk as unknowns. Then, the above relation becomes

$$(e(d_M M + \bar{d} L + d_L) - 1 + bk)(e(d_M M + \bar{d} L + d_L) - 1 + ak) - ak \cdot bk \cdot N = 0.$$

Let $x = \bar{d}$, $y = ak$, and $z = bk$. Then, the above equation can be written as a trivariate integer polynomial

$$f(x, y, z) = a_1 x^2 + a_2 xy + a_3 xz + a_4 yz + a_5 x + a_6 y + a_7 z + a_0 = 0, \quad (7)$$

with the coefficients given by

$$\begin{aligned} a_0 &= e^2 d_M^2 M^2 + 2e^2 d_M d_L M + e^2 d_L^2 - 2ed_M M - 2ed_L + 1, \\ a_1 &= e^2 L^2, \\ a_2 &= a_3 = eL, \\ a_4 &= -N + 1, \\ a_5 &= 2e^2 d_M M L + 2e^2 d_L L - 2eL, \\ a_6 &= a_7 = ed_M M + ed_L - 1. \end{aligned}$$

Thus, the task reduces to finding the small root $(x^*, y^*, z^*) = (\bar{d}, ak, bk)$ of the polynomial (7).

Using the estimates $e \simeq N^{1-\gamma}$ and $d \simeq N^\delta$, and from (4) we deduce that

$$ak \simeq N^{\delta-\gamma+1/2} \quad \text{and} \quad bk \simeq N^{\delta-\gamma+1/2}.$$

Thus, we set the upper bounds:

$$|x^*| \leq X = N^{\delta-\delta_M-\delta_L}, \quad |y^*| \leq Y = N^{\delta-\gamma+1/2}, \quad |z^*| \leq Z = N^{\delta-\gamma+1/2}. \quad (8)$$

In the lattice-based solving strategy, the maximal coefficient is defined by

$$U = \|f(xX, yY, zZ)\|_\infty,$$

which can be estimated as

$$\begin{aligned} U &= \max \left\{ |a_1|X^2, |a_2|XY, |a_3|XZ, |a_4|YZ, |a_5|X, |a_6|Y, |a_7|Z, |a_0| \right\} \\ &= \max \left\{ N^{2\delta-2\gamma-2\delta_M+2}, N^{2\delta-2\gamma-\delta_M+3/2}, N^{2\delta-2\gamma+2}, \right. \\ &\quad \left. N^{2\delta-2\gamma-\delta_M+2}, N^{2\delta-2\gamma+3/2}, N^{2\delta-2\gamma+2} \right\} \\ &= N^{2\delta-2\gamma+2}. \end{aligned}$$

Since X is smaller than both Y and Z , we apply an extra shift in the x -variable during the shift construction stage. For a chosen positive integer m and a non-negative integer $t = \tau m$, we define the monomial sets

$$\begin{aligned} \mathcal{M}_\uparrow &= \bigcup_{0 \leq l \leq t} \{x^{i+l}y^jz^k : x^i y^j z^k \in \text{monomial}(f^{m-1})\}, \\ \mathcal{M}_\downarrow &= \bigcup_{0 \leq l \leq t} \{x^{i+l}y^jz^k : x^i y^j z^k \in \text{monomial}(f^m)\}. \end{aligned}$$

That is, if we denote by $\mathcal{I}_\uparrow$ and $\mathcal{I}_\downarrow$ the corresponding integer index sets, then:

$$\begin{aligned} x^i y^j z^k \in \mathcal{M}_\uparrow &\iff \begin{cases} 0 \leq j \leq m-1, \\ 0 \leq k \leq m-1, \\ 0 \leq i \leq 2m-2+t-j-k; \end{cases} \\ x^i y^j z^k \in \mathcal{M}_\downarrow &\iff \begin{cases} 0 \leq j \leq m, \\ 0 \leq k \leq m, \\ 0 \leq i \leq 2m+t-j-k. \end{cases} \end{aligned}$$

Accordingly, the maximal degrees in $\mathcal{M}_\uparrow$ are $\mu_x = 2m-2+t$, $\mu_y = \mu_z = m-1$, and we set

$$R = U X^{2m-2+t} Y^{m-1} Z^{m-1}.$$

Then, the shift polynomials $g_{[i,j,k]}(x, y, z)$ are defined as

$$\begin{cases} x^i y^j z^k f(x, y, z) X^{2m-2+t-i} Y^{m-1-j} Z^{m-1-k}, & \text{if } x^i y^j z^k \in \mathcal{M}_\uparrow, \\ x^i y^j z^k R, & \text{if } x^i y^j z^k \in \mathcal{M}_\downarrow \setminus \mathcal{M}_\uparrow. \end{cases}$$

To be specific, the monomials of $\mathcal{M}_\downarrow \setminus \mathcal{M}_\uparrow$ are related to $[i, j, k]$ triples in $\mathcal{I}_\downarrow \setminus \mathcal{I}_\uparrow$ with

$$\begin{cases} j = m, \\ k = 0, \dots, m-1, \\ i = 0, \dots, m-2+t-k. \end{cases} \quad \begin{cases} j = 0, \dots, m-1, \\ k = m, \\ i = 0, \dots, m-2+t-j. \end{cases} \quad \begin{cases} j = m, \\ k = m, \\ i = 0, \dots, t. \end{cases}$$

and

$$\begin{cases} j = 0, \dots, m-1, \\ k = 0, \dots, m-1, \\ i = 2m-1+t-j-k, 2m+t-j-k. \end{cases}$$

The polynomial order $\prec_p$ (resp. the monomial order $\prec_m$) related to the row order (resp. the column order) of B are specified as follows. The rule is to arrange the shift polynomials $g_{[i,j,k]}$ in the ascending graded lexicographic order. The polynomial order $\prec_p$ is defined as $g_{[i,j,k]} \prec_p g_{[i',j',k']}$ if $i+j+k < i'+j'+k'$, or $i+j+k = i'+j'+k'$, $i > i'$, or $i+j+k = i'+j'+k'$, $i = i'$, $j > j'$. The monomial order $\prec_m$ is defined as $x^i y^j z^k \prec_m x^{i'} y^{j'} z^{k'}$ similarly. For instance, Table 1 provides a toy example of the lattice basis matrix B for $m = 1$ and $t = 1$, where “*” denotes a nonzero off-diagonal entry and other off-diagonal entries are 0.

Table 1. A toy example of lattice basis matrix B with $m = 1$ and $t = 1$.

$x^i y^j z^k$	1	x	y	z	x^2	xy	xz	yz	x^3	$x^2 y$	$x^2 z$	xyz
$g_{[0,0,0]}$	X	*	*	*	*	*	*	*	0	0	0	0
$g_{[1,0,0]}$	0	X	0	0	*	*	*	0	*	*	*	*
$g_{[0,1,0]}$	0	0	YR	0	0	0	0	0	0	0	0	0
$g_{[0,0,1]}$	0	0	0	ZR	0	0	0	0	0	0	0	0
$g_{[2,0,0]}$	0	0	0	0	$X^2 R$	0	0	0	0	0	0	0
$g_{[1,1,0]}$	0	0	0	0	0	XYR	0	0	0	0	0	0
$g_{[1,0,1]}$	0	0	0	0	0	0	XZR	0	0	0	0	0
$g_{[0,1,1]}$	0	0	0	0	0	0	0	YZR	0	0	0	0
$g_{[3,0,0]}$	0	0	0	0	0	0	0	0	$X^3 R$	0	0	0
$g_{[2,1,0]}$	0	0	0	0	0	0	0	0	0	$X^2 YR$	0	0
$g_{[2,0,1]}$	0	0	0	0	0	0	0	0	0	0	$X^2 ZR$	0
$g_{[1,1,1]}$	0	0	0	0	0	0	0	0	0	0	0	$XYZR$

The coefficient vectors of the scaled shift polynomials $g_{[i,j,k]}(xX, yY, zZ)$ are then used as the rows of a lattice basis matrix B regarding a lattice Λ . The respective diagonal elements of $g_{[i,j,k]}(xX, yY, zZ)$, i.e., the leading terms of

$g_{[i,j,k]}(xX, yY, zZ)$ are

$$\begin{cases} X^{2m-2+t}Y^{m-1}Z^{m-1}, & \text{if } x^i y^j z^k \in \mathcal{M}_\uparrow, \\ X^i Y^j Z^k R, & \text{if } x^i y^j z^k \in \mathcal{M}_\downarrow \setminus \mathcal{M}_\uparrow. \end{cases}$$

Since $R = UX^{2m-2+t}Y^{m-1}Z^{m-1}$, the determinant $\det(\Lambda)$ can be expressed (after appropriate grouping) in the form

$$\left(\frac{R}{U}\right)^{s_\uparrow} X^{s_X} Y^{s_Y} Z^{s_Z} R^{s_R},$$

where

$$s_\uparrow = |\mathcal{M}_\uparrow|, \quad s_R = |\mathcal{M}_\downarrow \setminus \mathcal{M}_\uparrow|, \quad \omega = |\mathcal{M}_\downarrow| = s_\uparrow + s_R,$$

and

$$s_X = \sum_{[i,j,k] \in \mathcal{I}_\downarrow \setminus \mathcal{I}_\uparrow} i, \quad s_Y = \sum_{[i,j,k] \in \mathcal{I}_\downarrow \setminus \mathcal{I}_\uparrow} j, \quad s_Z = \sum_{[i,j,k] \in \mathcal{I}_\downarrow \setminus \mathcal{I}_\uparrow} k.$$

Invoking the lattice condition (5) with $\omega = |\mathcal{M}_\downarrow| = |\mathcal{M}_\uparrow| + |\mathcal{M}_\downarrow \setminus \mathcal{M}_\uparrow| = s_\uparrow + s_R$ leads to

$$X^{s_X} Y^{s_Y} Z^{s_Z} < U^{s_\uparrow}. \quad (9)$$

By calculating $s_\uparrow$, s_X , s_Y , s_Z and representing them as functions of dominated terms through the above deduction, we obtain

$$\begin{aligned} s_\uparrow &= |\mathcal{M}_\uparrow| = \sum_{j=0}^{m-1} \sum_{k=0}^{m-1} \sum_{i=0}^{2m-2+t-j-k} 1 = m^3 + m^2 t + o(m^3), \\ s_X &= \sum_{[i,j,k] \in \mathcal{I}_\downarrow \setminus \mathcal{I}_\uparrow} i \\ &= \sum_{j=0}^m \sum_{k=0}^m \sum_{i=0}^{2m+t-j-k} i - \sum_{j=0}^{m-1} \sum_{k=0}^{m-1} \sum_{i=0}^{2m-2+t-j-k} i = \frac{7m^3}{3} + 3m^2 t + mt^2 + o(m^3), \\ s_Y &= \sum_{[i,j,k] \in \mathcal{I}_\downarrow \setminus \mathcal{I}_\uparrow} j \\ &= \sum_{j=0}^m \sum_{k=0}^m \sum_{i=0}^{2m+t-j-k} j - \sum_{j=0}^{m-1} \sum_{k=0}^{m-1} \sum_{i=0}^{2m-2+t-j-k} j = \frac{5m^3}{3} + \frac{3m^2 t}{2} + o(m^3), \\ s_Z &= \sum_{[i,j,k] \in \mathcal{I}_\downarrow \setminus \mathcal{I}_\uparrow} k \\ &= \sum_{j=0}^m \sum_{k=0}^m \sum_{i=0}^{2m+t-j-k} k - \sum_{j=0}^{m-1} \sum_{k=0}^{m-1} \sum_{i=0}^{2m-2+t-j-k} k = \frac{5m^3}{3} + \frac{3m^2 t}{2} + o(m^3). \end{aligned}$$

Letting m go to infinity with $t = \tau m$ (omitting its rounding) and ignoring lower-order terms $o(m^3)$ gives

$$s_{\uparrow} = (1 + \tau)m^3, \quad s_X = \left(\frac{7}{3} + 3\tau + \tau^2\right)m^3, \quad s_Y = s_Z = \left(\frac{5}{3} + \frac{3\tau}{2}\right)m^3. \quad (10)$$

Substituting the upper bounds from (8) and $U = N^{2\delta-2\gamma+2}$ with (10) into (9), and equating the exponents (with respect to N), we obtain the inequality

$$\begin{aligned} (\delta - \delta_M - \delta_L) \left(\frac{7}{3} + 3\tau + \tau^2\right) + 2 \left(\delta - \gamma + \frac{1}{2}\right) \left(\frac{5}{3} + \frac{3\tau}{2}\right) \\ < (2\delta - 2\gamma + 2)(1 + \tau). \end{aligned}$$

After further simplification, this reduces to

$$\delta \left(\frac{11}{3} + 4\tau + \tau^2\right) < (\delta_M + \delta_L) \left(\frac{7}{3} + 3\tau + \tau^2\right) + \gamma \left(\frac{4}{3} + \tau\right) + \frac{1}{3} + \frac{\tau}{2}.$$

Rearranging yields

$$\delta < \delta_M + \delta_L + \frac{8\gamma - 8\delta_M - 8\delta_L + 2 + (6\gamma - 6\delta_M - 6\delta_L + 3)\tau}{22 + 24\tau + 6\tau^2}. \quad (11)$$

Optimizing the right-hand side with respect to τ , define

$$\eta := \gamma - \delta_M - \delta_L \quad \text{and} \quad \xi := \frac{8\eta + 2 + (6\eta + 3)\tau}{22 + 24\tau + 6\tau^2}.$$

It can be shown that the maximal value of ξ is achieved at

$$\tau_0 = \frac{\sqrt{4\eta^2 + 20\eta + 13} - 8\eta - 2}{3(2\eta + 1)},$$

which leads to

$$\xi \leq \eta + 1 - \frac{\sqrt{4\eta^2 + 20\eta + 13}}{4}.$$

Substituting back $\eta = \gamma - \delta_M - \delta_L$, we deduce that

$$\delta < \gamma + 1 - \frac{\sqrt{4(\gamma - \delta_M - \delta_L)^2 + 20(\gamma - \delta_M - \delta_L) + 13}}{4}, \quad (12)$$

which holds when $\gamma \leq \delta_M + \delta_L + \frac{3}{10}$ (so that $\tau_0 \geq 0$ and hence $\eta \leq \frac{3}{10}$).

For the case $\gamma > \delta_M + \delta_L + \frac{3}{10}$, one should set $\tau_0 = 0$, and (11) then directly implies

$$\delta < \frac{4\gamma + 7\delta_M + 7\delta_L + 1}{11}. \quad (13)$$

Collecting (12) and (13) yields the final attack bound:

$$\begin{cases} \delta < \gamma + 1 - \frac{\sqrt{4(\gamma - \delta_M - \delta_L)^2 + 20(\gamma - \delta_M - \delta_L) + 13}}{4}, & \gamma \leq \delta_M + \delta_L + \frac{3}{10}, \\ \delta < \frac{4\gamma + 7\delta_M + 7\delta_L + 1}{11}, & \gamma > \delta_M + \delta_L + \frac{3}{10}. \end{cases}$$

Once the integer polynomial (7) is solved, we can recover the common root $(x^*, y^*, z^*) = (\bar{d}, ak, bk)$. Since $\gcd(a, b) = 1$, we can compute

$$k = \gcd(y^*, z^*),$$

from which it follows that $a = y^*/k$ and $b = z^*/k$. With the recovered $\bar{d}$ (and hence d), the equation $ed = 2gabk + 1$ (with known a , b , and k) yields

$$g = \frac{ed - 1}{2abk}.$$

Finally, the primes are given by

$$p = 2ga + 1, \quad q = 2gb + 1,$$

which furnishes the factorization of N . All these steps, including lattice reduction and root extraction, are executable in time polynomial in ℓ . $\square$

Remark 1. It is interesting to note that when $\delta_M = \delta_L = 0$, our proposed attack reduces to the small private key attack on common prime RSA as described in [14] (i.e., in the absence of any partial key exposure). In contrast, when a fraction of the most or least significant bits is leaked, our attack extends and strengthens the small private key attack, thereby enlarging the vulnerable bound on the private key.

Remark 2. Our attack would be easier to understand if defining $\delta = \delta_U + \delta_K$ with expressions $\delta_U = \delta - \delta_M - \delta_L$ and $\delta_K = \delta_M + \delta_L$. This makes the dependence on known (i.e., δ_K) vs. unknown (i.e., δ_U) bits explicit. One may readily see that the attack bound on δ increases as the attacker knows more bits, and that it is irrelevant whether MSBs or LSBs are known.

Remark 3. The bounds of Proposition 2 can be reproduced from the small private key attack on common prime RSA in [35, Proposition 1] by substituting $\delta' = \delta - \delta_K$ and $\gamma' = \gamma - \delta_K$.

4 Experimental Results

In order to assess the practicality and effectiveness of our proposed partial key exposure attack on common prime RSA (as formalized in Proposition 2), we carried out a series of numerical experiments. All experiments were performed using SageMath [30] on a system configured with a 64-bit Windows 10 platform (running Ubuntu 22.04 via WSL 2). The common prime RSA instances were generated with parameters chosen at random.

Initially, we generated RSA moduli of bit-length ℓ (equaling 1024, 2048, 4096 to demonstrate practicality on modern key sizes) by varying the parameter γ (which governs the size of g). For each instance, a private exponent d with a predetermined bit-length was produced, with specified segments of its most

significant bits (MSBs) and/or least significant bits (LSBs) exposed. The corresponding public exponent e was then computed from the relation $ed \equiv 1 \pmod{\text{lcm}(p-1, q-1)}$. By gradually increasing the bit-length of d , we were able to achieve larger values of δ that fall within the vulnerable range predicted by our theoretical analysis. The open-source implementation of the proposed attacks is available at https://github.com/MengceZheng/CPRSA_PKEA.

To implement the attack, we selected appropriate lattice parameters m and t to construct a lattice tailored for our instance. Table 2 summarizes the experimental outcomes. In this table, the γ column specifies the size of g , while $\delta_M \ell$ and $\delta_L \ell$ denote the number of exposed bits in the MSB and LSB portions, respectively. The column δ_t lists the theoretical upper bound for δ , and δ_e shows its experimentally determined value. The lattice parameters m and t determine the lattice dimension, given in the ω column, and the overall computational time, comprising both the LLL reduction and subsequent Gröbner basis computations, is recorded in the Time column (in seconds).

Table 2. Experimental results of the proposed partial key exposure attack

ℓ	γ	$\delta_M \ell$	$\delta_L \ell$	δ_t	δ_e	m	t	ω	Time
1024	0.10	51	0	0.164	0.083	2	0	27	0.93 s
1024	0.15	92	0	0.207	0.124	2	0	27	0.95 s
1024	0.20	122	0	0.244	0.166	2	0	27	0.91 s
1024	0.25	153	0	0.280	0.209	2	0	27	0.96 s
2048	0.17	200	0	0.219	0.145	2	0	27	2.39 s
2048	0.22	250	0	0.252	0.164	2	0	27	2.24 s
4096	0.20	491	0	0.244	0.166	2	0	27	7.82 s
1024	0.15	0	40	0.174	0.112	2	0	27	0.92 s
1024	0.20	0	102	0.230	0.169	2	0	27	0.86 s
1024	0.25	0	163	0.287	0.224	3	0	64	36.39 s
1024	0.30	0	225	0.344	0.272	3	1	80	182.25 s
2048	0.17	0	200	0.219	0.145	2	0	27	2.21 s
2048	0.22	0	250	0.252	0.164	2	0	27	2.17 s
4096	0.25	0	655	0.287	0.211	2	0	27	7.43 s
1024	0.25	92	51	0.274	0.208	2	0	27	1.06 s
1024	0.30	102	102	0.330	0.249	2	0	27	0.96 s
1024	0.35	153	81	0.367	0.287	3	0	64	35.72 s
1024	0.40	40	256	0.424	0.333	3	0	64	37.46 s
2048	0.35	280	150	0.353	0.225	2	0	27	2.15 s
2048	0.40	100	400	0.393	0.291	3	0	64	123.23 s
4096	0.25	327	327	0.287	0.205	2	0	27	7.68 s

For each experiment, we generated a sufficient number of shift polynomials whose corresponding integer polynomial equations shared a common root. As observed in Table 2, the computational time grows with the lattice dimension, which is largely influenced by both the dimension itself and the magnitude of the matrix entries. The derived vectors were then converted into integer polynomials that satisfy the conditions for successful root recovery.

The root extraction process was implemented as follows. After obtaining the reduced vectors, we recovered the unknown component $\bar{d}$ and the products ak and bk . These values were derived from a Gröbner basis algorithm for extracting the common root. With $\bar{d}$, ak , and bk in hand, the factorization of the RSA modulus N was completed by computing $k = \gcd(ak, bk)$, determining a and b , and finally recovering g via the key equation. Although the recovered experimental bounds were below the theoretical insecure bound (primarily due to hardware limitations), we expect that further improvements in lattice construction (e.g., using higher-dimensional lattices) can bridge this gap.

To illustrate the procedure concretely, we present a toy example with a small ℓ below.

Example 1. Consider a toy instance in which we set $\gamma = 0.1$ and use two 256-bit balanced primes, so that the composite modulus N has a bit-length of $\ell = 512$. Suppose the private exponent d is 79 bits long, and that 20 MSBs and 25 LSBs of d are known. The specific parameters for this instance are:

$$\begin{aligned} N &= 944015881214494062765132743122430079279877027987760987450204 \\ &\quad 610805878114116507679725280979682982258595993690865594139268 \\ &\quad 3630718642688098595947826056605595, \\ e &= 158077612337460119529619367564438184550521081977907875063017 \\ &\quad 572198240897628237047803359799839950943707585996554330167537 \\ &\quad 25293112516788607, \\ d_M &= 585670 \text{ (i.e., } 10001110111111000110), \\ d_L &= 856799747 \text{ (i.e., } 1111010010001100110001111). \end{aligned}$$

In this example, we derived parameters a_i for $0 \leq i \leq 7$ and then generated the target trivariate integer polynomial $f(x, y, z)$. By setting $m = 3$ and $t = 1$, we constructed a lattice of dimension $\omega = 80$. After approximately 56 seconds of computation, the desired root (x^*, y^*, z^*) was successfully extracted:

$$\begin{aligned} x^* &= 7600455243, \\ y^* &= 470719931192695442899091272918982120932401559897714908818003 \\ &\quad 95173348660509862017480, \\ z^* &= 640980693440577964751321756304968619219723208475194389458837 \\ &\quad 40230806466640514667820. \end{aligned}$$

Using y^* and z^* , we computed $k = \gcd(y^*, z^*)$ as

$$k = 2908377569207397435020,$$

and hence

$$a = 16184966359817508199533906423147448480943318821053062250566374,$$

$$b = 22039115561438625728027143037476181391585030854211092058691641.$$

Finally, we derived the common prime g as

$$g = 2572211484787583,$$

which enabled the factorization of N into

$$p = 832623127032465501913388602186294860608849934831056191226302 \\ 93267053665068085,$$

$$q = 113378532323386346820067590302927355093573441389249628217599 \\ 951023731565387407.$$

One may readily verify that $N = pq$, confirming the validity of the partial key exposure attack as dictated by Proposition 2. We provide more toy numerical examples when only MSBs or LSBs is leaked in Appendix B. One may produce additional numerical examples for common prime RSA modulus of larger bit-length via https://github.com/MengceZheng/CPRSA_PKEA.

5 Concluding Remarks

In this work, we introduced a new partial key exposure attack on common prime RSA that employs a lattice-based strategy to solve trivariate integer polynomial equations. By generalizing the traditional small private key attack, our approach accommodates attack scenarios in which only a fraction of the private key's bits (from both its most and least significant parts) is exposed. We derived the attack bounds under which the RSA modulus can be factored in polynomial time and provided a detailed demonstration.

Our experimental results demonstrate that even partial leakage of key bits can significantly compromise the security of common prime RSA. Although the overall efficiency is influenced by the lattice dimension and the specific parameters chosen for the lattice construction, the experiments affirm the theoretical predictions and the practical viability of the proposed attack.

Future work mainly includes interpolating smoothly between common prime variant and standard RSA by analyzing the case $\gamma = \delta_M = \delta_L = 0$ and closing the gap to Boneh-Durfee's bound $\delta < 0.292$.

Acknowledgments. Mengce Zheng gratefully acknowledges the financial support for academic visiting provided by the China Scholarship Council (CSC No. 202308330390). This work was partially supported by the National Natural Science Foundation of China (No. 62002335), and Ningbo Young Science and Technology Talent Cultivation Program (No. 2023QL007).

Disclosure of Interests. The authors have no competing interests.

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A Algorithm

Algorithm 1: Common Prime Generation

Input: Modulus bit-length n and γ
Output: Common primes p and q

```

1  $g \leftarrow$  a random  $[\gamma n]$ -bit prime;
2 while  $p, q, h$  are not primes do
3   while  $\gcd(a, b) \neq 1$  do
4      $a, b \leftarrow$  two random  $[(1/2 - \gamma)n - 1]$ -bit positive integers;
5   end
6    $p \leftarrow 2ga + 1$ ;
7    $q \leftarrow 2gb + 1$ ;
8    $h \leftarrow 2gab + a + b$ ;
9 end
10 return  $p$  and  $q$ 

```

B Numerical Examples

B.1 Leaked MSBs

We provide a toy instance with $\ell = 512$, $\gamma = 0.1$, $\delta = 0.078$, $\delta_M = 0.05$, and $\delta_L = 0$. The related parameters for this instance are as follows.

$N = 107374379172778170604042854989720956959647738164874913130348$
 $313845323603599601631006597693952857463036116336113359267089$
 $17283884604777889614028444748003231,$

$e = 637379973709147474567304848439405742212285325607597220554721$
 $356145991235454863607446598092930881588878289065446292523154$
 $365624952198774619,$

$d_M = 23771926$ (i.e., $1011010101011101100010110$),

$d_L = 0$.

By setting $m = 2$ and $t = 0$, we constructed a lattice of dimension $\omega = 27$. After approximately one second of computation, the desired root (x^*, y^*, z^*) was successfully extracted:

$x^* = 22319,$

$y^* = 453636671427988818041955360188029006589199434541165041486855$
 $8350955140946,$

$z^* = 506077047772035316087132283751430055048494040424060822364528$
 $1919443543507.$

Using y^* and z^* , we computed $k = \gcd(y^*, z^*) = 255411358841$, and hence

$$\begin{aligned} a &= 17761021807584879761849391686664739699661078729304916829591906, \\ b &= 19814195033004817029766827031554325070178566687783844128102827. \end{aligned}$$

Finally, we derived the common prime $g = 2761837656493103$, which enabled the factorization of N into

$$\begin{aligned} p &= 981061176919662409577869782522625034776630816245938538211430 \\ &\quad 07014069187248637, \\ q &= 109447179950502611031119740174429093796449080955017755341750 \\ &\quad 304679506800604363. \end{aligned}$$

B.2 Leaked LSBs

We provide a toy instance with $\ell = 512$, $\gamma = 0.1$, $\delta = 0.082$, $\delta_M = 0$, and $\delta_L = 0.03$. The related parameters for this instance are as follows.

$$\begin{aligned} N &= 100511134556907920205886186234170183083389524761386816285803 \\ &\quad 465276827502242735438637605442242644783371299106548836013355 \\ &\quad 71366090682251104126216944003749205, \\ e &= 939353451070463440666724732224461227006258618924391042566347 \\ &\quad 250666553619499610836879518284179388118468048772462756432466 \\ &\quad 655875310607167825, \\ d_M &= 0, \\ d_L &= 27707 \text{ (i.e., } 110110000111011). \end{aligned}$$

By setting $m = 2$ and $t = 0$, we constructed a lattice of dimension $\omega = 27$. After approximately one second of computation, the desired root (x^*, y^*, z^*) was successfully extracted:

$$\begin{aligned} x^* &= 52225948, \\ y^* &= 148886615324013021690440390921758665516437622959400329473668 \\ &\quad 14846114761143, \\ z^* &= 172687453809197567305330305264439883985342705994066565401811 \\ &\quad 78673022718021. \end{aligned}$$

Using y^* and z^* , we computed $k = \gcd(y^*, z^*) = 1130627741929$, and hence

$$\begin{aligned} a &= 15273590714709511940560013141994661344172016216848559949431549, \\ b &= 13168491255131669189272721387561910514451765453377764628171167. \end{aligned}$$

Finally, we derived the common prime $g = 3534585632619791$, which enabled the factorization of N into

$$\begin{aligned} p &= 930903199873355130961795238490079233527386452164111737702521 \\ &\quad 74863062759532195, \\ q &= 107971628597454572042992767201154273826212457211419268840416 \\ &\quad 370965044194372519. \end{aligned}$$